



MOTOR CONTROL THROUGH PWM

■ SUNIL KUMAR

Motion control plays a vital role in industrial automation. It is required in manufacturing plants of industries like chemical,

control through pulse-width modulation (PWM).

As shown in the circuit, the 230V, 50Hz AC mains is stepped down by transformer X1 to deliver a secondary output of 12V, 1A. The transformer

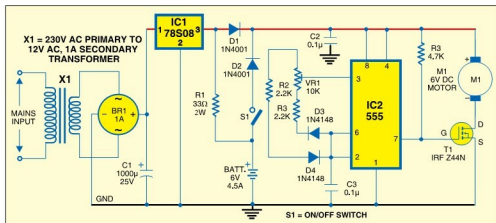
is connected to the gate of MOSFET T1. Potentiometer VR1 controls the charge and discharging time of capacitor C3 and thus decides the duty cycle of

PWM. In other words, using VR1 you can control the speed of the DC motor. The DC motor is connected between the positive and drain terminals of MOSFET T1.

Working of the circuit is simple. Connect the DC motor as shown in the circuit. Now vary potentiometer VR1 slowly. The speed of the motor will vary accordingly, provided mains or the battery is providing

power to the circuit.

Assemble the complete circuit on a general-purpose PCB and house in a small cabinet. Fix switch S1 on the front panel of the cabinet. Fix a two-pin connector at the rear side of the panel for connecting the motor to be controlled and also connect the mains power cord for input mains supply. ●



pharmaceutical, plastic and textile. And it may be a flat-belt application, flow-control application or mixing of substances. Different types of motors—AC, DC, servo or stepper—are used depending upon the application. Of these, DC motors are widely used because controlling a DC motor is somewhat easier than other kinds of motors. Here we describe DC motor

output is rectified by bridge rectifier BR1, filtered by capacitor C1 and regulated by IC 78S08 (IC1). Capacitor C2 bypasses the ripples present in the regulated supply. When mains is available, the battery charges through resistor R1. It provides power to the circuit when mains fails. PWM is done using timer 555, whose pin 7 is used as the output and con-

MOVIE MAKER

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Like real movies, this circuit makes use of a characteristic of the human eye and brain known as the persistence of vision. A sequence of still pictures is projected onto a screen in rapid succession. The pictures differ slightly from one another and the brain interprets the succession of still pictures as continuous motion.

Here the pictures are shadows cast by low-voltage lamps. There are four

lamps in all, which glow in sequence cyclically. This gives the illusion of a simple but realistic movie.

Fig. 1 shows the circuit for the movie maker. It is driven by clock pulses provided by NAND gates N1 and N2. The flickering frequency is adjustable through preset VR1. A suitable rate for perceiving continuous motion is 16 Hz. The clock pulses are fed to counter IC CD4022 (IC2). IC2 has eight outputs, but only the first four

(0-3) are used in this circuit. The outputs go high one at a time, in sequence. The fifth output (output 4) is connected to the reset input so that the counter is immediately reset at the fifth count and the first output (output 0) goes high.

The counter outputs are fed to CD4049 hex buffer (IC3). The buffer outputs drive transistors T1 through T4 in a sequence. As each transistor conducts, the lamp connected to it glows. The lamps are rated at 0.3A so these provide enough light to operate the movie show in a dimly-lit room.

Assemble the circuit on a general-purpose PCB. Power-on the circuit

